KCPC-CS120
Problem Sheet
Combinatorics I

Developed by the KCPC Internal Committee, this problem sheet accompanies the *Combinatorics I* workshop. Problems are divided into three categories:

- Integrated examples (discussed live)
- Core practice set (to be attempted during the workshop)
- Extended homework

I. Integrated Into Slides (Live Walkthrough)

1. **Project Euler 15 - Lattice Paths**

   https://projecteuler.net/problem=15

2. **LeetCode 62 - Unique Paths**

   https://leetcode.com/problems/unique-paths/description/

3. **Project Euler 53 - Combinatoric Selections**

   https://projecteuler.net/problem=53

4. **Project Euler 24 - Lexicographic Permutations**

   https://projecteuler.net/problem=24

II. Core Workshop Practice Set

These problems should be attempted during the workshop. They require careful modelling and correct algorithm selection.

For any questions that were not completed during the workshop, please attempt them independently afterwards.

1. **CodeForces 1743A - Password**

   https://codeforces.com/problemset/problem/1743/A

2. **CodeForces 1821A - Matching**

   https://codeforces.com/problemset/problem/1821/A

3. **Project Euler 31 - Coin Sums**

   https://projecteuler.net/problem=31

4. **Codeforces C - Good Subarrays**

   https://codeforces.com/problemset/problem/1398/C

5. **AtCoder D - Knight** Very good! It is interesting, will require some knowledge of number theory https://atcoder.jp/contests/abc145/tasks/abc145_d

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III. Extended Homework

Important. The following problems are genuinely extremely difficult. They should only be attempted after completing all previously listed problems and assigned supplementary reading. They are intended as stretch challenges for those who wish to explore advanced modelling and optimisation techniques.

Yersin's Note.

I hope you enjoy working through these problems.

If you have any questions, please feel free to contact me or the Director of Competitive Programmes (Marcell).

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1. **Project Euler 483 - Repeated Permutation**

   https://projecteuler.net/problem=483

2. **Leetcode 22 - Generate Parantheses**

   https://leetcode.com/problems/generate-parentheses/description/

3. **AtCoder T - Permutation**

   https://atcoder.jp/contests/dp/tasks/dp_t

Guidance:

- Identify whether order matters: determine whether to use permutations or combinations.
- When direct counting overcounts, consider inclusion-exclusion or division by symmetry.
- For hard problems, try to find a bijection to a set that is easier to count.
- Focus on thinking of a nice argument before commiting anything to code.